Louis ALMAIRAC

Master MVA student at ENS Paris-Saclay and engineering student at École des Ponts, working on machine learning for time series and biosignals.

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EDUCATION

École Normale Supérieure Paris-Saclay • *Paris, France* *2026 – 2027*
Master MVA (Mathematics, Vision, Learning)
Relevant courses: Time Series, Object Recognition, Foundations of LLMs, Computational Statistics, Convex Optimisation

École Nationale des Ponts et Chaussées • *Paris, France* *2022 – 2027*
Master of Engineering: Mathematics and Computer Science
Relevant courses: Machine Learning, Stochastic Processes, Deep Learning, Statistics, Computer Vision, Programming

Lycée Pierre de Fermat • *Toulouse, France* *2020 – 2022*
Intensive Mathematics, Physics and Computer Science courses preparing for France's top engineering schools.

EXPERIENCE

Research Intern, Université du Québec à Montréal • *Montréal, Canada* *March 2026 – July 2026*

- Decoded in-game attentional engagement from EEG with Prof. Mounir Boukadoum in collaboration with Ubisoft; first-author paper submitted to IEEE ISCAS 2027.
- Designed a pipeline combining multivariate empirical mode decomposition, time-frequency analysis and Riemannian geometry, on par with an end-to-end EEG-Conformer under leave-one-subject-out evaluation.
- Ran control analyses (ICA, eye-tracking-verified labels, permutation tests, self-supervised probes) showing that the decodable signal is a play-vs-rest arousal shift, not a within-gameplay marker.

Machine Learning Intern, Caisse des Dépôts • *Paris, France* *June 2025 – Dec. 2025*

- Implemented a LambdaMART learning-to-rank model selecting countries within an emerging-market equity universe, replacing a neural ListMLE baseline and optimising NDCG directly.
- Engineered the financial features (momentum, volatility, FX-equity correlation, drawdowns) and a purged, embargoed temporal cross-validation ruling out look-ahead bias.

Medical Time-Series Research Project, Ponts et Chaussées • *Paris, France* *Nov. 2024 – June 2025*

- Developed a time-frequency Granger causality framework combining wavelet decomposition and VAR modelling to analyse EEG time series during sleep, highlighting state-dependent brain interactions.
- Presented a poster at the 4th International Summer Institute on Network Physiology, Lake Como School of Advanced Studies, Italy, 2025.

Data Scientist Intern, Sopra Steria • *Paris, France* *Jan. 2024 – June 2024*

- Cleaned inconsistent operational data and built the analysis tools identifying the causes of metro traffic delays, for an RATP line-automation project.

PROJECTS & COMPETITIONS

Martha, AI Stylist App (personal project) *2025 – 2026*
Designing and shipping a native iOS app (SwiftUI, CoreData, Supabase) that digitises a wardrobe and recommends outfits with LLMs and vision models, from data model and backend to interface.

Medical Data Challenge, QRT *2025*
Built survival prediction models for blood cancer patients from clinical and molecular data, evaluated with IPCW-C-index, in collaboration with Institut Gustave Roussy.

Hackathons, Mistral AI x Alan & Nuit pour Entreprendre *2024*
Ranked 4th out of 100+ participants with an LLM-powered food-waste prevention app built in 24h.
Grand Prix of Nuit pour Entreprendre with Blue Seed, an AI irrigation-planning solution for farmers pitched in 10h.

Mathematics Competition, TFJM^2 *2020 and 2023*
Achieved 2nd place nationally in the 2020 edition as part of an 8-member team; Community Manager of the 2023 edition.

SKILLS & INTERESTS

Languages	French (native), English (fluent), Spanish (fluent)
Programming	Advanced skills in Python (PyTorch, scikit-learn, pandas)
Hobbies	Competitive Rugby player, Member of English Debating Club